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# Gap Image Classification Based on PCA-SVM with Multiple Feature Fusion

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***Abstract*—The detection of gap size at the threaded connection of the inner diameter of an aviation rocket body is crucial for the performance of the body. Due to the different processing degrees of the inner diameter during processing, there are two working conditions on the inner diameter surface. The image features of the gap under these two working conditions are different. Therefore, before detecting the gap size, it is necessary to distinguish between these two working conditions. This article proposes a multi feature fusion classification method based on PCA-SVM, which takes texture features, grayscale features, and depth features as joint feature parameters of the image. The PCA algorithm is used to perform multi feature fusion on them, and the fused feature parameters are input into the training SVM classifier for training the classification network. The experiment shows that the proposed method has a classification recognition rate of 99.5% in the classification of rough and fine surfaces of aviation rockets, Providing technical support for efficient and automated production of products.**

***Keywords*—gap, pca, svm, multi-feature fusion, image classification***

## I. INTRODUCTION

Due to the complex surface morphology and narrow space inside the rocket projectile, relying solely on manual detection not only has low efficiency but also poor reliability. Therefore, it is necessary to introduce machine vision technology.

S Majumdar et al[1]. utilized machine vision technology to extract the morphological, texture, and color features of grains, thereby achieving classification and recognition of wheat, barley, and rye. KM Rajpot et al[2] generated energy functions from various wavelet subbands based on two-dimensional discrete wavelet transform and inputted them as feature vectors into SVM to achieve texture image classification. Experiments have shown that their proposed method has good classification results for both single class and multi class texture images. Ahmed Mohiuddin applied the convolutional neural network (CNN) to the medical field [3], and converted the pathological image of breast cancer to be classified into four images with different resolutions in parallel. At the same time, he extracted the features of these four images through EfficientNet-B0 network, so as to classify the fused features. Experiments show that the method is far better than previous research in accuracy, precision, recall, etc.

In summary, this project draws on the research foundation of the aforementioned literature to study the recognition and classification of surface images of the inner diameter of the projectile body. The specific process of the algorithm in this article is shown in the figure 1.

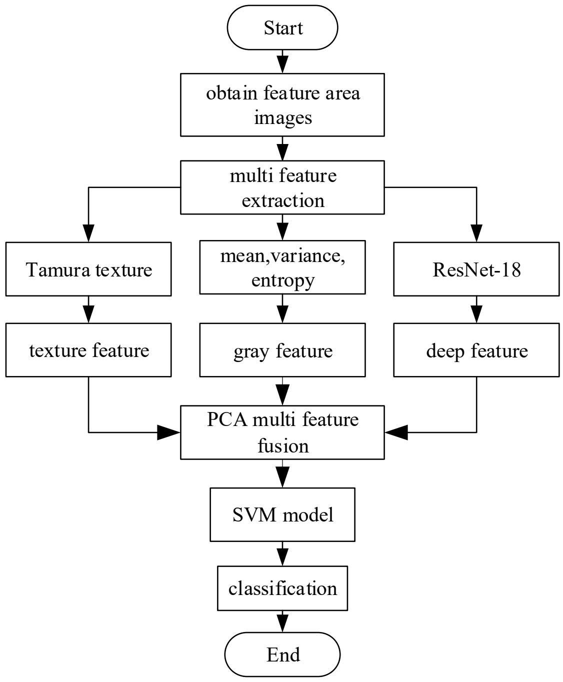


Fig. 1. Flow chart of multi feature fusion classification algorithm based on PCA-SVM

## II. MULTI FEATURE EXTRACTION

The feature extraction of images is a crucial step in the image processing process, especially in image classification. Selecting different features for analysis based on the different characteristics of the image is the key, and different image features represent different image information. The more effective information obtained, the more advantageous the final image classification will be.

Figure 2 shows the ROI images of the inner diameter surface under two working conditions after being collected and processed by an industrial camera. It is found that the surface features of the left area of the two working conditions images are different. After analyzing the location of the gap area, Select the image in the red box area in the following figure as the ROI area for this article.

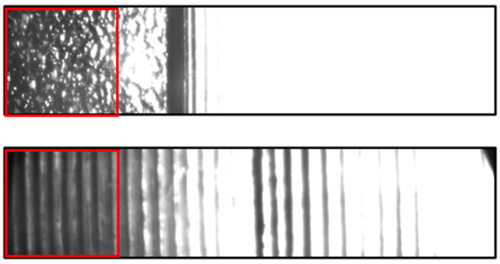

Fig. 2. Example of a figure caption

## A. Texture Feature Extraction

Tamura texture [4] is proposed based on the psychological research of human visual perception of texture, consisting of six feature components, namely roughness, contrast, directio- nality, linearity, regularity, and coarseness. Next, we will introduce these six attributes separately.

(1)Roughness: Roughness is the most essential feature in a texture. In the roughness calculation of Tamura texture, the first step is to center each pixel $(x,y)$ in the feature area, and calculate the average gray value $G_k(x,y)$ of all pixels in the window with size $2^k$.

$$G_k(x,y) = \sum_{i=x-2^{k-1}}^{x+2^{k-1}} \sum_{j=y-2^{k-1}}^{y+2^{k-1}} f(i,j) / 2^{2k} \tag{1}$$

In the equation(1) , the value of k is taken as 0,1,2,3,4,5; $f(i,j)$ is the pixel grayscale value of the position $(i,j)$ in the pixel coordinate system.

Then calculate the average grayscale difference between each pixel's non overlapping windows in the horizontal and vertical directions $E_{k,\mathrm{h}}(x,y)$ and $E_{k,\mathrm{h}}(x,y)$.in equation (2),(3).

$$E_{k,h}(x,y) = | G_k(x+2^{k-1},y) - G_k(x-2^{k-1},y) | \tag{2}$$

$$E_{k,v}(x,y) = | G_k(x,y+2^{k-1}) - G_k(x,y-2^{k-1}) | \tag{3}$$

Finally, calculate the window size at which the average grayscale difference at each pixel point is the highest, and use the average value of $S_{\text{best}}$ at all points in the image as the roughness parameter $T_{\text{coa}}$ , as shown in equation (4):

$$T_{\text{coa}} = \frac{1}{col * row} \sum_{i=1}^{col} \sum_{j=1}^{row} S_{\text{best}}(i,j) \tag{4}$$

In the above equation, *col* and *row* represent the width and height of the feature area respectively.

(2)Contrast: In the Tamura texture, the calculation process of the contrast parameter , as shown in equation (5):

$$T_{\text{con}} = \sigma^2 \mu_4^{-\frac{1}{4}} \tag{5}$$

In the above formula, $\sigma$ is the standard deviation of the image, and $\mu_4$ is the fourth order central moment of gray scale.

(3)Directionality: Directionality represents the global texture characteristics of an image. In the calculation of Tamura texture directionality parameters, the convolutional template shown in equation (6) is first constructed to obtain the gradient amplitude $\Delta G(x,y)$ and directional angle of each pixel $\theta(x,y)$,as shown in equation (7):

$$\begin{cases} |\Delta G(x,y)| = (|\Delta_H| + |\Delta_V|)/2 \\ \theta(x,y) = \tan^{-1}(\dfrac{\Delta_V}{\Delta_H}) + \dfrac{\pi}{2} \end{cases} \tag{6}$$

$$\Delta H = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix} \qquad \Delta V = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \tag{7}$$

Construct a directional angle histogram $H_D(k)$ by counting the number of pixels whose gradient amplitude and directional angle are greater than the pre-set threshold in all pixels of the image.

$$H_D(k) = N_\theta(k) / \sum_{i=0}^{n-1} N_\theta(i) \tag{8}$$

In equation(8), $N_\theta(k)$ is the number of pixels which $|\Delta G(x,y)| \geq t$ and $(2k-1)\pi/2n \leq \theta \leq (2k+1)\pi/2n$, and t is the pre-set threshold.

$$T_{\text{dir}} = 1 - rn_p \sum_{p}^{n_p} \sum_{\phi \in w_p} (\phi - \phi_p)^2 (\phi) H_D \tag{9}$$

In the formula (9), $p$ is the peak value of the histogram, $n_p$ is the number of all peaks in the histogram. For each peak $p$, $W_p$ represents the range of $\theta$ values contained in the peak, $\phi$ is the maximum peak in the histogram $w_p.r$ is the histogram normalization factor.

(4) Linearity: the definition of the Tamura texture linearity parameter is based on the size m×m of the window, the average degree of overlap in a common edge direction occurs in a pair of pixels with a distance of $d$ along the gradient direction of a certain pixel. In order to simplify the direction, divide it into eight directions: 0°, 45°, 90°,135°,180°,225°, 270°,and 315°,the specific definition is shown in equation (10):

$$T_{\text{lin}} = \frac{\sum_{i=1}^{m} \sum_{j=1}^{m} P_a(i,j) \cos[(i-j)\frac{2\pi}{m}]}{\sum_{i=1}^{m} \sum_{j=1}^{m} P_a(i,j)} \tag{10}$$

In the equation, $P_a(i,j)$ is the distance point of the local direction co-occurrence matrix in the window of $m \times m$.

(5)Regularity: Regularity represents the degree of regularity of a texture in an image, which is established based on the description of the image by human psychology, the more regular the texture of the image, the larger the regularity parameter. as shown in equation (11):

$$T_{\text{reg}} = 1 - r(\sigma_{\text{coa}} + \sigma_{\text{con}} + \sigma_{\text{dir}} + \sigma_{\text{lin}}) \tag{11}$$

In the formula, r is the parameter normalization factor, and $\sigma_{\text{coa}}$, $\sigma_{\text{con}}$, $\sigma_{\text{dir}}$, and $\sigma_{\text{lin}}$ are the standard deviations of their texture parameters.

(6)Roughness: In Tamura texture, roughness parameters are related to roughness parameters and contrast parameters, as shown in the following equation(12):

$$T_{\text{rou}} = T_{\text{coa}} + T_{con} \tag{12}$$

### B. Grayscale Feature Extraction

The grayscale feature is used to describe the local or global grayscale properties of an image, which can be obtained by statistical analysis of the image grayscale histogram or direct operation of its global grayscale value. The commonly used grayscale feature parameters include grayscale mean, grayscale variance, and grayscale entropy.

(1)Grayscale mean: as a global grayscale feature, grayscale mean represents the average grayscale value of all pixels in an image, reflecting the overall brightness and darkness of the image, the larger the value, the brighter the overall image, as shown in equation (13):

$$G_a = \frac{1}{col*row}\sum_{i=1}^{col}\sum_{j=1}^{row} f(i,j) \tag{13}$$

(2)Grayscale variance: The grayscale variance of an image represents the average degree of overall grayscale change in the image. The larger the grayscale variance of the image, the higher the clarity of the image; as shown in equation (14):

$$G_v = \frac{1}{col*row}\sum_{i=1}^{col}\sum_{j=1}^{row} (f(i,j) - G_a)^2 \tag{14}$$

(3) Gray scale entropy: entropy is a statistic, and the one-dimensional entropy value of an image reflects the amount of information contained in the image gray scale distribution.

$$G_e = \sum_{i=0}^{255} p_i \log p_i \tag{15}$$

The $p_i$ in the formula(15) represents the probability of a certain grayscale value appearing in the image.

### C. Depth Feature Extraction

In order to extract the more essential feature information of the image, this paper uses the convolution neural network to independently learn the image to obtain information as the depth feature of the image. Residual network (ResNet) [5], as a mature convolutional neural network model, is widely used in the field of image classification. It proposes residual learning during the convolution process, using residual blocks as the basic structure of the model, alleviating the problem of model degradation caused by excessive layers, and achieving good results in the classification field.

This article uses the ResNet-18 network pre trained on the ImageNet dataset to obtain model parameters as the feature extraction model parameters, as the inner diameter gap feature area image is a single channel grayscale image, three identical single channel images are concatenated and used as model input. Then, the feature data $f_i$ obtained from the fully connected layer, namely the FC layer, is weighted and summed, and normalized to obtain four depth features, as the final depth feature vector $D$, as shown in equation (16):

$$D_i = \sum_{j=128i}^{128(i+1)} f_j / 128, (i = 0,1,2,3) \tag{16}$$

## III. Multi Feature Classification

After multiple feature extraction of the image, feature fusion is required to classify the fused feature data. A PCA-SVM image classification algorithm is proposed. Firstly, principal component analysis is used to reduce the dimensionality of joint features, ensuring the completeness of image feature information while reducing data dimensionality; Then, the dimensionality reduced feature data is concatenated and input into a support vector machine for classification.

### A. Multi feature fusion

By extracting different features of an image, more complete information can be obtained, but at the same time, the dimensionality of image features is also increasing, and there may be some redundant or even irrelevant feature information in the extracted multi feature parameters, which has a negative impact on image classification work.

Principal Component Analysis (PCA) [6] is a commonly used multivariate analysis method in machine learning. Its main goal is to compress data to reduce feature dimensions while minimizing information loss. The basic idea of PCA is to search for a set of orthogonal vectors in the feature space of the original sample, and map and transform the original n-dimensional data onto the k-dimensional space based on the maximum variance. The retained k-dimensional information is the principal component features of the original data, thus achieving the use of low-dimensional data to describe the original high-dimensional data while maximizing the preservation of the information of the original data. The specific introduction of PCA algorithm is as follows:

(1)First, sample $i$ samples, each with $j$ feature data $m_{ij}$, and after sampling an $i{\times}j$ completed characteristic matrix M of is as shown in equation (17):

$$M = \begin{bmatrix} M_1 \\ M_2 \\ \vdots \\ M_i \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1j} \\ m_{21} & m_{22} & \cdots & m_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ m_{i1} & m_{i2} & \cdots & m_{ij} \end{bmatrix}_{i\times j} \tag{17}$$

(2) Centralize the feature vector $M_i$ of each sample, which means that the data of all feature vectors is averaged to obtain a new feature matrix, and calculate the covariance matrix $C$ of the sample data after centralization. as shown in equation (18):

$$C = \frac{1}{i-1}\hat{M}\hat{M}^T \tag{18}$$

In order to rank the contribution of data features and obtain the first $k$-dimensional principal components, the eigenvalues and eigenvectors of the covariance matrix $C$ are obtained. The eigenvalues are sorted by size and the first $k$ eigenvalues and their corresponding eigenvectors are taken, as shown in equation (19):

$$Cv_k = \lambda_k v_k \tag{19}$$

### *B. SVM classification*

Support Vector Machine(SVM) is a supervised machine learning classification method , which outperforms traditional classification algorithms in terms of generalization ability. The main principle is to find a hyperplane that can separate all data in the original or high dimensional new feature space to achieve the purpose of classification , as shown in figure 3. SVM classifier can effectively solve learning problems in small sample data, and has its own advantages in nonlinear and high-dimensional data classification.

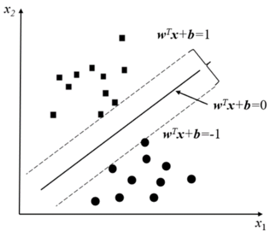


Fig. 3. Example of a figure caption

Since the multi-feature parameters extracted by this algorithm belong to high-dimensional, nonlinear separable data, they cannot be directly distinguished by linear function. When SVM deals with this classification problem, it first maps the linear indivisible data of the original space to the high-dimensional space, but the algorithm complexity of finding the inner product through vectors in this process is very large, so it introduces the kernel function $K(x_i, x_j)$, as shown in equation (20) , uses the kernel function calculation to replace the inner product of the nonlinear mapping function, and converts it into a linear divisible problem in the high-dimensional space.

$$K(x_i, x_j) = \phi(x_i)\phi(x_j) \tag{20}$$

## IV. EXPERIMENT

This experiment selected 100 feature area image samples from both precision and rough machined surfaces, extracted feature parameters from all sample images, and established feature datasets for multi feature fusion and classification experiments. Firstly, PCA feature dimensionality reduction is performed on these 13 features. Figure 4 shows the contribution rate curve of each component obtained after PCA dimensionality reduction. When the number of principal components $k$=6, the cumulative contribution rate is 94.62%; When the number of principal components $k$=7, the cumulative contribution rate reaches 97.81%. Therefore, in this algorithm, the number of principal components after dimensionality reduction is set to 7.

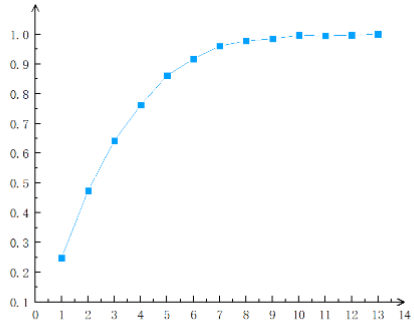


Fig. 4. Example of a figure caption

This article uses the grid method to study the radial basis of Gaussian kernel functions σ , and the penalty coefficient $C$ is designed to obtain the optimal parameters through experiments. Through experimental testing, when the radial basis function in this algorithm σ When the penalty coefficient $C$=4 is 0.43, the classification accuracy of our model is the highest.

This article compares the recognition accuracy of single feature SVM classification method, SVM classification method without multi feature fusion, and BP classification algorithm after multi feature fusion with the algorithm in this article, as shown in the Table 1, it can be found that the classification performance of multi feature extraction is significantly improved compared to single feature extraction methods.

TABLE I. TABLE TYPE STYLES

| Classification Method | Accuracy |
|---|---|
| texture Features - SVM | 87.5% |
| grayscale features - SVM | 83.7% |
| depth features - SVM | 85.4% |
| multi-features - SVM | 94.4% |
| multi-features fusion – PCA-SVM | 99.5% |

## V. CONCLUSION

This article studies the classification algorithm for surface images of internal diameter gaps of projectile bodies. In terms of feature extraction, a multi feature extraction method was proposed, which obtained more original information of the image. In terms of feature classification, a multi feature fusion classification algorithm based on PCA-SVM is proposed, which uses Principal Component Analysis (PCA) for feature fusion, reducing the original 13 dimensional feature data to 7 dimensions. Input the fused features into a Support Vector Machine (SVM) classifier for binary image classification. The experimental results show that the classification algorithm proposed in this paper achieves a classification accuracy of 99.5%

in the dataset constructed in this paper, which can achieve accurate classification of missile body gap images.

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